	SAMRAT ASHOK TECHNOLOGICAL INSTITUTE (DEGREE) VIDISHA (M.P)
Department	Electronics Department
Branch	Electronics & Instrumentation Engineering
Year & Sem	B.Tech. First Year (I - Semester/ II- Semester)
Session	Batch 2026-27 onwards

Category of Course	Course Title	Course Code	Credits - 3			Theory Paper
			L	T	P	
ECS	Fundamentals of Electronics Engineering	EI26101	3	0	0	Max.Marks - 70 Min.Marks - 22 Duration - 3 Hrs.

Course Code	Course Title	Maximum Marks Allotted						Total Marks
		Theory Paper			Practical			
		End Sem.	Mid Sem. MST	Quiz/ Assignment	End Sem.	Lab Work	Assign/Quiz /Term paper	
EI26101	Fundamentals of Electronics Engineering	70	20	10	--	--	--	100

Course Objectives	To impart knowledge of basic operation of semiconductor devices, converters, amplifiers and applications.
Prerequisite Knowledge	Basic knowledge of Semiconductor Physics and Mathematics.
Course Description	Solid background of fundamentals of electronics and exposure to state art technologies and its control aspects used in practice.
Course Outcomes	**Upon completion of the course, student will be able to:** 1. Discuss and demonstrate the principles of different semiconductor diodes and their applications. 2. Familiarize different types of BJT transistors with their configurations and biasing techniques. Apply these biasing techniques for transistor amplification. 3. Understand the construction, operation, and characteristics of FETs and MOSFETs, and apply this knowledge to analyze and design basic amplifier and switching circuits in electronics. 4. Analyze multistage and tuned amplifiers and their applications. 5. To gather knowledge about power semiconductor devices, applications and wave generators.

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	0	0	0	0	0	0	0	0	0	0	0	1	0	0
CO2	3	0	0	0	0	0	0	0	0	0	0	0	3	0	0
CO3	3	2	1	0	0	0	0	0	0	0	0	0	2	0	0
CO4	0	2	0	0	0	0	0	0	0	0	0	0	3	0	0
CO5	3	0	2	0	0	0	0	0	0	0	0	0	3	0	0

(0 = Not correlated; 1 = weakly correlated; 2 = Moderately correlated; 3 = strongly correlated)

Syllabus

Unit - I	**Semiconductor Diodes:** P-N junction diode, Rectifiers, Clipping and Clamping, RC Filters, Zener diode, Zener Regulator, Tunnel diode, Varactor diode, Light Emitting Diode (LED), PIN diode, P-N-P-N diode	**CO1**
Unit - II	**Bipolar Junction Transistors (BJTs):** Physical structure and operation modes, Transistor as an amplifier, Basic BJT amplifier configuration: common emitter, common base and common collector amplifiers, Biasing the BJT: fixed bias, emitter feedback bias, collector feedback bias and voltage divider bias, Transistor as a switch: cut-off and saturation modes.	**CO2**
Unit – III	**Field Effect Transistors:** working principle, types and transfer characteristics. JFET amplifier and applications. MOSFET, its types, characteristics and applications. CMOS basics.	**CO3**
Unit - IV	**Multistage Amplifiers:** Classification of multi-stage amplifier, types of coupling, Darlington amplifier, classification of power amplifiers and its types, , concept of tuned amplifiers.	**CO4**
Unit – V	**Power Electronics and Wave Generators:** Basic principle of power devices, characteristics of SCR, Diac, Triac, device specifications, turn-on and turn-off methods, Controlled rectifiers and inverters. **Wave Generators:** Principle, Classification, popular Oscillators and Multivibrators.	**CO5**

Textbooks:

1. Electronic Devices and Circuits, 11th edition – Boyelstad and Nashelsky – PHI Publisher
2. Electronic Devices and Circuits, 5th edition – David A. Bell – PHI Publisher
3. Principles of Electronics - V.K. Mehta and Rohit Mehta - S. Chand Publisher

Reference Books:

1. Electronic Principles, 9th edition – A.P. Malvino *et al.* - Tata McGraw Hill Publisher
2. Basic Electrical and Electronics Engineering, 2nd edition - D.P. Kothari and I.J Nagrath - McGraw Hill Publisher
3. Integrated Electronics - Millman Halkias, TMH publisher

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Category of Course	**Course Title**	**Course Code**				
			L	**T**	**P**	**Credits - 1**
DLC	Fundamentals of Electronics Engineering	EI26101P	0	0	2	

Course Code	Course Title	**Maximum Marks Allotted**					**Total Marks**
		Theory Paper			Practical		
		End Sem.	Mid Sem. MST	Quiz/ Assignment	End Sem.	Lab Work	
EI26101P	Fundamentals of Electronics Engineering	--	--	--	30	20	50

Course Objectives	To impart practical aspect of the basic operation of semiconductor devices, converters, amplifiers and applications.
Prerequisite Knowledge	Basic knowledge of Semiconductor Physics and operation of basic electronic equipments.
Course Description	Solid background of fundamentals of electronics and exposure to state art technologies and its control aspects used in practice.
Course Outcomes	**Upon completion of the course, student will be able to:** 1. Demonstrate and plot the V-I Characteristics of basic semiconductor diodes and understand its applications. 2. Familiarize with different characteristics of BJT transistors with their configurations. 3. Plot and analyze the transfer characteristics of FET device 4. Analyze multistage amplifier and plot the graphical performance parameters. 5. Application of power devices to obtain different types of outputs.

List of Experiments

1.	To draw the forward and reverse bias characteristics of a semiconductor P-N junction diode.	CO1
2.	Use the Zener diode as a voltage regulator and draw the regulation characteristics..	CO1
3.	To observe the output waveform of a half wave rectifier. Calculate ripple factor and efficiency.	CO1
4.	To observe the output waveform of full wave rectifier. Calculate PIV, ripple factor and rectification efficiency.	CO1
5.	To observe the waveform of positive and negative Clamper circuit.	CO1
6.	To observe the waveform of Clipper circuit in different positions.	CO1
7.	To plot common base input and output characteristics for P-N-P bipolar junction transistor.	CO2
8.	To plot common emitter input and output characteristics for N-P-N bipolar junction transistor.	CO2
9.	To plot the transfer characteristics of an FET device	CO3
10.	To draw the frequency response of two stage RC coupled amplifier.	CO4
11.	To observe the output waveform of a Silicon controlled rectifier (SCR) and find out the delay angle.	CO5